

Statistical Modeling (MT-2002)

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Course Instructors

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Final Exam

Total Time (Hrs): 3

Total Marks: 120

Total Questions: 6

Roll No

Section

Student Signature

Attempt all the questions

Use the answer sheet for any calculation required. Your answer for each question should be on this sheet. Use the bubble sheet to answer the MCQs for Question 6.

DO NOT WRITE BELOW THIS LINE

CLO # 1: Use random variables to represent uncertainty.

Question # 1:

[15 Marks]

A university administration wants to study the uncertainty in the number of emails sent by professors to the Campus Director during a semester. Let the random variable Y_i represent the number of emails sent by the i^{th} professor.

The dataset contains four independent variables (AdminRole, Experience, ResearchLoad, Meetings) and one response variable (Emails).

- Emails: Number of emails sent to the Director
- AdminRole: 1 if professor has administrative role; 0 otherwise
- Experience: Years of teaching experience
- ResearchLoad: Number of research projects
- Meetings: Number of meetings with Director

AdminRole	Experience	ResearchLoad	Meetings	Emails
0	2	1	0	10
1	10	4	6	12
0	3	2	0	100
1	15	5	8	18
0	1	1	0	11
1	12	3	5	0
0	4	2	0	10

Questions:

1. Assume that the response variable *Emails* represents count data where the non-zero counts are approximately evenly distributed, meaning no specific count value occurs substantially more frequently than others. Which Generalized Linear Model (GLM) would be most appropriate for modeling this response variable? Justify your choice based on the nature of the distribution.

Your answer below this line

Poisson Regression is the most appropriate GLM because the response variable *Emails* is count data and the counts are assumed to follow a standard count-generating process without excessive zeros.

2. Suppose further investigation reveals that nearly 80% of the observations in *Emails* are zero counts. Which probabilistic distribution would now be more suitable for representing the uncertainty in the response variable? Write distribution name and justify your reasoning in one line.

Your answer below this line

Zero-Inflated Poisson (ZIP) Distribution. It is suitable because nearly 80% of the observations are zeros, which indicates excess zeros beyond what a standard Poisson distribution can model.

3. Write the selected model(s) in proper statistical notation:

Your answer below this line

Poisson Regression Model

$$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 10)$$

$$\mu_i = \exp(\beta_0 + \beta_1 \text{AdminRole}_i + \beta_2 \text{Experience}_i + \beta_3 \text{ResearchLoad}_i + \beta_4 \text{Meetings}_i)$$

$$\text{Emails}_i \sim \text{Poisson}(\mu_i)$$

Zero-Inflated Poisson (ZIP) Model

$$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 10)$$

$$\psi \sim \text{Beta}(1, 1)$$

$$\mu_i = \exp(\beta_0 + \beta_1 \text{AdminRole}_i + \beta_2 \text{Experience}_i + \beta_3 \text{ResearchLoad}_i + \beta_4 \text{Meetings}_i)$$

$$\text{Emails}_i \sim \text{ZIP}(\psi, \mu_i)$$

CLO # 1: Use random variables to represent uncertainty.

Question # 2 :

[10 Marks]

In a logistic regression model, the relationship between predictor variable x and probability p is given by:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

Suppose the estimated model is:

$$\log\left(\frac{p}{1-p}\right) = -2 + 0.8x$$

Answer the following:

1. [2 marks] Calculate the **log-odds** when $x = 3$.

Your answer below this line

$$\begin{aligned}\log\left(\frac{p}{1-p}\right) &= -2 + 0.8(3) \\ &= -2 + 2.4 = 0.4\end{aligned}$$

Therefore, the log-odds is 0.4.

2. [2 marks] Convert the log-odds into **odds**.

Your answer below this line

The odds are obtained by exponentiating the log-odds:

$$\text{Odds} = e^{0.4}$$

$$\text{Odds} \approx 1.492$$

Therefore, the odds are 1.492.

3. [2 marks] Calculate the corresponding **probability** p .

Your answer below this line

Using

$$p = \frac{\text{Odds}}{1 + \text{Odds}}$$

$$p = \frac{1.492}{1 + 1.492}$$

$$p = \frac{1.492}{2.492} \approx 0.599$$

Therefore, the probability is

$$p \approx 0.599$$

or approximately 59.9%.

4. [2 marks] Interpret the meaning of coefficient $\beta_1 = 0.8$ in terms of log-odds and likelihood of success.

Your answer below this line

For every one-unit increase in x , the log-odds of success increase by 0.8.

5. [2 marks] Explain what would happen to the probability if the coefficient becomes negative instead of positive.

Your answer below this line

If the coefficient is negative, increasing x would decrease the log-odds and the odds of success. As a result, the probability of success would decrease as x increases. Therefore, larger values of x would be associated with a lower likelihood of the event occurring.

CLO # 3: Perform simulation studies to evaluate stochastic models.

Question # 3:

[25 + 5 = 30 Marks]

Consider the observed dataset $x = [1, 2, 3, 4, 5]$, $y = [3, 5, 7, 9, 11]$, and the test statistic $T(y) = \frac{1}{n} \sum_{i=1}^n y_i$. Assume the following Bayesian Linear Regression model: $y_i = \alpha + \beta x_i$. Five candidate models are provided, where each model contains five posterior samples of the parameters (α, β) .

Posterior Samples of Candidate Models

Model	Sample	α	β	Notation
M_1	1	1	2	$M_1^{(1)}$
	2	2	2	$M_1^{(2)}$
	3	1	2.5	$M_1^{(3)}$
	4	2	2.5	$M_1^{(4)}$
	5	1	3	$M_1^{(5)}$
M_2	1	0	2	$M_2^{(1)}$
	2	1	2	$M_2^{(2)}$
	3	0	2.2	$M_2^{(3)}$
	4	1	2.2	$M_2^{(4)}$
	5	0	2.5	$M_2^{(5)}$
M_3	1	2	1.5	$M_3^{(1)}$
	2	3	1.5	$M_3^{(2)}$
	3	2	1.8	$M_3^{(3)}$
	4	3	1.8	$M_3^{(4)}$
	5	2	2	$M_3^{(5)}$

Model	Sample	α	β	Notation
M_4	1	0	1	$M_4^{(1)}$
	2	1	1	$M_4^{(2)}$
	3	0	1.2	$M_4^{(3)}$
	4	1	1.2	$M_4^{(4)}$
	5	0	1.5	$M_4^{(5)}$
M_5	1	1	1.8	$M_5^{(1)}$
	2	2	1.8	$M_5^{(2)}$
	3	1	2	$M_5^{(3)}$
	4	2	2	$M_5^{(4)}$
	5	1	2.2	$M_5^{(5)}$

Replicated datasets are generated using

$$y_i^{rep} = \alpha + \beta x_i$$

The Bayesian p -value is defined as

$$p_B = P(T(y^{rep}) \geq T(y))$$

Questions

1. Compute the observed statistic $T(y)$.
2. Compute the replicated dataset y^{rep} for each posterior sample.
3. Compute $T(y^{rep})$ for all replicated datasets.
4. Compute the Bayesian p -value for each model.
5. Determine the best fitting model.

Working Table

Posterior Sample	Replicated Dataset y^{rep}	$T(y^{rep})$
$M_1^{(1)}$	[3, 5, 7, 9, 11]	7.0
$M_1^{(2)}$	[4, 6, 8, 10, 12]	8.0
$M_1^{(3)}$	[3.5, 6, 8.5, 11, 13.5]	8.5
$M_1^{(4)}$	[4.5, 7, 9.5, 12, 14.5]	9.5
$M_1^{(5)}$	[4, 7, 10, 13, 16]	10.0
$M_2^{(1)}$	[2, 4, 6, 8, 10]	6.0
$M_2^{(2)}$	[3, 5, 7, 9, 11]	7.0
$M_2^{(3)}$	[2.2, 4.4, 6.6, 8.8, 11]	6.6
$M_2^{(4)}$	[3.2, 5.4, 7.6, 9.8, 12]	7.6
$M_2^{(5)}$	[2.5, 5, 7.5, 10, 12.5]	7.5
$M_3^{(1)}$	[3.5, 5, 6.5, 8, 9.5]	6.5
$M_3^{(2)}$	[4.5, 6, 7.5, 9, 10.5]	7.5
$M_3^{(3)}$	[3.8, 5.6, 7.4, 9.2, 11]	7.4

Posterior Sample	Replicated Dataset y^{rep}	$T(y^{rep})$
$M_3^{(4)}$	[4.8, 6.6, 8.4, 10.2, 12]	8.4
$M_3^{(5)}$	[4, 6, 8, 10, 12]	8.0
$M_4^{(1)}$	[1, 2, 3, 4, 5]	3.0
$M_4^{(2)}$	[2, 3, 4, 5, 6]	4.0
$M_4^{(3)}$	[1.2, 2.4, 3.6, 4.8, 6]	3.6
$M_4^{(4)}$	[2.2, 3.4, 4.6, 5.8, 7]	4.6
$M_4^{(5)}$	[1.5, 3, 4.5, 6, 7.5]	4.5
$M_5^{(1)}$	[2.8, 4.6, 6.4, 8.2, 10]	6.4
$M_5^{(2)}$	[3.8, 5.6, 7.4, 9.2, 11]	7.4
$M_5^{(3)}$	[3, 5, 7, 9, 11]	7.0
$M_5^{(4)}$	[4, 6, 8, 10, 12]	8.0
$M_5^{(5)}$	[3.2, 5.4, 7.6, 9.8, 12]	7.6

$$p_B(M_1) = 1.0 \quad p_B(M_2) = 0.6 \quad p_B(M_3) = 0.8$$

$$p_B(M_4) = 0.0 \quad p_B(M_5) = 0.8$$

Best Fitting Model: M_2

CLO # 3: Perform simulation studies to evaluate stochastic models.

Question # 4 :

[15 Marks]

Consider the statistical models model₁, model₂, model₃, model₄, and model₅. The Bayes Factor (ratio of two marginal likelihoods) values between different candidate models are provided below.

Note: Justify all answers using appropriate evidence strength.

Bayes Factor	Comparison	Value
BF ₁	$\frac{p(y \text{model}_2)}{p(y \text{model}_1)}$	12
BF ₂	$\frac{p(y \text{model}_3)}{p(y \text{model}_1)}$	55
BF ₃	$\frac{p(y \text{model}_4)}{p(y \text{model}_1)}$	0.8
BF ₄	$\frac{p(y \text{model}_5)}{p(y \text{model}_1)}$	120
BF ₅	$\frac{p(y \text{model}_5)}{p(y \text{model}_3)}$	2.1

- (i) [2 marks] Based on BF_1 , which model is better: model₂ or model₁?

Your answer below this line

Answer: model₂ is preferred because $BF_1 = 12$ indicates strong evidence in favor of model₂.

- (ii) [2 marks] Based on BF_2 , which model is better: model₃ or model₁?

Your answer below this line

Answer: model₃ is preferred because $BF_2 = 55$ indicates very strong evidence in favor of model₃.

- (iii) [2 marks] Based on BF_3 , which model is preferred: model₄ or model₁?

Your answer below this line

Answer: model₁ is preferred because $BF_3 = 0.8 < 1$, indicating evidence against model₄.

- (iv) [2 marks] Based on BF_4 , which model is better: model₅ or model₁?

Your answer below this line

Answer: model₅ is preferred because $BF_4 = 120$ indicates decisive evidence in favor of model₅.

- (v) [2 marks] Compare model₅ and model₃ using BF_5 . Which model is preferred?

Your answer below this line

Answer: model₅ is slightly preferred because $BF_5 = 2.1$ indicates weak to moderate evidence in favor of model₅

- (vi) [2 marks] Based on all Bayes Factor values, determine the overall best fitting model among all candidate models.

Your answer below this line

Bayes Factor	Preferred Model
$BF_1 = 12$	model_2 over model_1
$BF_2 = 55$	model_3 over model_1
$BF_3 = 0.8$	model_1 over model_4
$BF_4 = 120$	model_5 over model_1
$BF_5 = 2.1$	model_5 over model_3

Overall Best Fitting Model = model_5

Reason: Model_5 is preferred over both model_1 and model_3 and has the strongest Bayes Factor evidence among all candidate models.

- (vii) [3 marks] Consider a scenario with two datasets (one large and one small), two competing models (Model A and Model B), and two prior settings (strong priors and weak priors). Model selection must be performed strictly using either Information Criteria

(WAIC/LOO) or Bayes Factors, which approach is expected to provide more stable and reliable model selection across different dataset sizes and prior specifications? your answer should be information criteria or bayes factor?

Your answer below this line

Information Criteria (WAIC/LOO)

Justification: Information Criteria (WAIC/LOO) are generally more stable and reliable across different dataset sizes and prior specifications because they focus on predictive accuracy. In contrast, Bayes Factors are sensitive to prior choices and can produce different conclusions under strong and weak priors. Therefore, WAIC/LOO is the preferred approach for consistent model selection.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 5 :

[15 Marks]

Consider the following table returned by the `arviz.compare()` function, then answer the following question.

Model	Rank	elpd_waic	p_waic	elpd_diff	weight	se	dse	warning	scale
model_2	0	0.0761	0.0014	0.0000	0.2038	0.0085	0.0000	False	negative_log
model_0	1	0.0819	0.0012	0.0058	0.2027	0.0093	0.0129	False	negative_log
model_1	2	0.1358	0.0037	0.0597	0.1924	0.0329	0.0350	False	negative_log
model_3	3	0.3991	0.0451	0.3230	0.1486	0.0865	0.0909	False	negative_log
model_4	4	0.7565	0.0453	0.1034	0.0787	0.0738	0.0737	False	negative_log

Table 3: WAIC-based comparison of candidate models

1. [2 marks] Identify the best and worst model from the predictive accuracy point of view only. Write the model names and justify your selection using the appropriate column(s) of the table.

Your answer below this line

The best model is **model_2** because it has Rank = 0 and the smallest elpd_waic and elpd_diff (0.0000). The worst model is **model_4** because it has the lowest rank (Rank = 4) and the largest elpd_waic & elpd_diff value (0.7565).

2. [2 marks] Which model has the highest uncertainty in predictive performance estimation? Justify your answer using the appropriate column of the table.

Your answer below this line

model_3 has the highest uncertainty because it has the largest standard error value, $se = 0.0865$.

3. [2 marks] Based on the `warning` column, can we conclude that WAIC estimation is reliable for all models? Justify your answer.

Your answer below this line

Yes. The WAIC estimation appears reliable for all models because the `warning` column is `False` for every model, indicating no reliability issues were detected.

4. [2 marks] Which model has the smallest effective number of parameters? Justify your answer using the appropriate column.

Your answer below this line

`model_0` has the smallest effective number of parameters because it has the lowest `p_waic` value, 0.0012.

5. [2 marks] Compare `model_2` and `model_0` using the `elpd_diff` column. What can you conclude regarding their predictive performances?

Your answer below this line

The predictive performances of `model_2` and `model_0` are almost identical because the difference in `elpd` values is very small (`elpd_diff` = 0.0058).

6. [2 marks] Observe the table given above and select a model that best from predictive accuracy as well as parameter complexity overall. Justify your selections from table.

Your answer below this line

`model_0` is the best overall choice because it has predictive accuracy very close to `model_2` (`elpd_diff` = 0.0058) while having the smallest effective number of parameters (`p_waic` = 0.0012). Thus, it provides a better balance between accuracy and simplicity.

7. [3 marks] Using the results from posterior predictive checks (Question #3), Bayes Factors (Question #4), and WAIC-based predictive accuracy analysis (Question #5), determine the overall best fitting model among all candidate models. Justify your answer by combining evidence from all three model comparison approaches.

Your answer below this line

Method	Selected Model
Posterior Predictive Check	M_1
Bayes Factors	M_5
WAIC	<code>model_2</code>

Justification: The three model comparison approaches do not agree completely. Posterior predictive checks favor M_1 , Bayes Factors favor M_5 , and WAIC favors `model_2`. Hence no single model is best.

CLO # 1: Use random variables to represent uncertainty.

Question # 6 :

[35 Marks]

Consider the following Bayesian models in PyMC, next answer the objective type questions, choose the right options for each

Model 1

```
1 with pm.Model() as model_1:
2     p = pm.Beta('p', alpha=1, beta=1)
3
4     y = pm.Bernoulli('heads', p=p, observed=y)
5
6     trace = pm.sample(1000, return_inferencedata=True)
```

Model 2

```
1 with pm.Model() as model_2:
2     mu = pm.Uniform('μ', lower=40, upper=70)
3
4     sigma = pm.HalfNormal('σ', sigma=10)
5
6     y = pm.Normal('y', mu=μ, sigma=σ, observed=y)
7
8     idata_g = pm.sample(1000)
```

Model 3

```
1 with pm.Model() as Model_3:
2     alpha = pm.Normal('α', mu=0, sd=10)
3     beta = pm.Normal('β', mu=0, sd=1)
4     epsilon = pm.HalfNormal('ε', 5)
5
6     mu = pm.Deterministic('μ', alpha + beta * x)
7
8     y_pred = pm.Normal('y_pred', mu=μ, sd=epsilon, observed=y)
9
10    idata_g = pm.sample(2000, tune=2000,
11                        return_inferencedata=True)
```

Model 4

```
1 with pm.Model() as Model_4:
2     alpha = pm.Normal('α', mu=0, sd=10)
3     beta = pm.Normal('β', mu=0, sd=10)
4
5     mu = alpha + pm.math.dot(X_c, beta)
6
7     theta = pm.Deterministic('θ', pm.math.sigmoid(mu))
8
9     bd = pm.Deterministic('bd', -alpha/beta)
10
11    y1 = pm.Bernoulli('y1', p=theta, observed=y)
12
13    idata_theta = pm.sample(1000, return_inferencedata=True)
```

Model 5

```
1 with pm.Model() as model_5:
2     alpha_tmp = pm.Normal('α_tmp', mu=0, sd=10)
3
4     beta = pm.Normal('β', mu=0, sd=1, shape=2)
5
6     epsilon = pm.HalfCauchy('ε', 5)
7
8     mu = alpha_tmp + pm.math.dot(X_centered, beta)
9
10    alpha = pm.Deterministic('α', alpha_tmp - pm.math.dot(X_mean,
11                                                         beta))
12
13    y_pred = pm.Normal('y_pred', mu=μ, sd=epsilon, observed=y)
14
15    idata = pm.sample(2000, return_inferencedata=True)
```

Model 6

```
1 with pm.Model() as model_6:
2     alpha = pm.Normal('α', mu=0, sd=10)
3
4     beta = pm.Normal('β', mu=0, sd=2, shape=len(x_n))
5
6     mu = alpha + pm.math.dot(x_1, beta)
7
8     theta = pm.Deterministic('θ', 1 / (1 + pm.math.exp(-mu)))
9
10    bd = pm.Deterministic('bd', -alpha/beta[1] - beta[0]/beta[1] * x_1
11                           [:,0])
12
13    y1 = pm.Bernoulli('y1', p=theta, observed=y)
14
15    idata_1 = pm.sample(2000, target_accept=0.9,
16                       return_inferencedata=True)
```

Model 7

```
1 with pm.Model() as model_7:
2     alpha = pm.Normal('α', mu=y_2.mean(), sd=1)
3
4     beta1 = pm.Normal('β1', mu=0, sd=1)
5
6     beta2 = pm.Normal('β2', mu=0, sd=1)
7
8     epsilon = pm.HalfCauchy('ε', 5)
9
10    mu = alpha + beta1 * x_2 + beta2 * x_2**2
11
12    y_pred = pm.Normal('y_pred', mu=μ, sd=epsilon, observed=y)
13
14    idata = pm.sample(2000, return_inferencedata=True)
```

Model 8

```
1 with pm.Model() as Model_8:
2     psi = pm.Beta('ψ', 1, 1)
3
4     alpha = pm.Normal('α', 0, 10)
5
6     beta = pm.Normal('β', 0, 10, shape=2)
7
8     theta = pm.math.exp(alpha + beta[0] * data['x1'] + beta[1] * data['x2'])
9
10    y1 = pm.ZeroInflatedPoisson('y1', psi, theta, observed=y)
11
12    idata = pm.sample(1000, return_inferencedata=True)
13
14    az.plot_trace(idata)
```

Model 9

```
1 with pm.Model() as model_9:
2     μ = pm.Uniform('μ', 40, 75)
3
4     σ = pm.HalfNormal('σ', sigma=10)
5
6     ν = pm.Exponential('ν', 1/30)
7
8     y = pm.StudentT('y', mu=μ, sigma=σ, nu=ν, observed=y)
9
10    idata_t = pm.sample(1000)
```

Model 10

```
1 with pm.Model() as model_10:
2     μ = pm.Beta('μ', 1, 1)
3
4     κ = pm.HalfNormal('κ', 1000)
5
6     θ = pm.Beta('θ', alpha=μ*κ, beta=(1.0-μ)*κ, shape=3)
7
8     y = pm.Bernoulli('y', p=θ[group_idx], observed=y)
9
10    idata_h = pm.sample(2000)
```

1. In Model 1, which prior distribution is assigned to parameter p ?
 - (A) Normal
 - (B) Beta
 - (C) Poisson
 - (D) All of the Above
 - (E) None of the Above
2. Which model is appropriate for estimating probabilities of binary outcomes?
 - (A) Model 1
 - (B) Model 2
 - (C) Model 7
 - (D) All of the Above
 - (E) None of the Above
3. In Model 2, the parameter σ represents:
 - (A) Mean
 - (B) Standard deviation
 - (C) Probability
 - (D) All of the Above
 - (E) None of the Above
4. Which distribution is used as the likelihood in Model 2?
 - (A) Bernoulli
 - (B) Poisson
 - (C) Normal
 - (D) All of the Above
 - (E) None of the Above
5. Which model performs Bayesian linear regression?
 - (A) Model 3
 - (B) Model 4
 - (C) Model 8
 - (D) All of the Above
 - (E) None of the Above
6. In Model 3, $\mu = \alpha + \beta x$ represents:

- (A) Prior
 - (B) Linear predictor
 - (C) Posterior
 - (D) All of the Above
 - (E) None of the Above
7. Which model is designed for binary classification problems?
- (A) Model 4
 - (B) Model 2
 - (C) Model 7
 - (D) All of the Above
 - (E) None of the Above
8. In logistic regression, the sigmoid function converts values into:
- (A) Counts
 - (B) Probabilities
 - (C) Standard deviations
 - (D) All of the Above
 - (E) None of the Above
9. Which model uses multiple predictors for continuous response modeling?
- (A) Model 5
 - (B) Model 1
 - (C) Model 8
 - (D) All of the Above
 - (E) None of the Above
10. In Model 5, the keyword `shape=2` indicates:
- (A) Two observations
 - (B) Two predictors
 - (C) Two priors
 - (D) All of the Above
 - (E) None of the Above
11. Which model performs multiple logistic regression?
- (A) Model 6
 - (B) Model 3
 - (C) Model 2
 - (D) All of the Above
 - (E) None of the Above
12. In Model 6, θ represents:
- (A) Probability of class membership
 - (B) Mean of response

- (C) Standard deviation
 - (D) All of the Above
 - (E) None of the Above
13. Which model captures nonlinear relationships using polynomial terms?
- (A) Model 7
 - (B) Model 1
 - (C) Model 4
 - (D) All of the Above
 - (E) None of the Above
14. In Model 7, the term $\beta_2 x^2$ helps model:
- (A) Curvature
 - (B) Missing values
 - (C) Sampling bias
 - (D) All of the Above
 - (E) None of the Above
15. Which model is suitable for count data with many zero observations?
- (A) Model 8
 - (B) Model 3
 - (C) Model 5
 - (D) All of the Above
 - (E) None of the Above
16. In Model 8, ψ controls:
- (A) Probability of extra zeros
 - (B) Mean of data
 - (C) Number of predictors
 - (D) All of the Above
 - (E) None of the Above
17. Which model is robust to outliers?
- (A) Model 9
 - (B) Model 2
 - (C) Model 4
 - (D) All of the Above
 - (E) None of the Above
18. In Model 9, parameter ν controls:
- (A) Degrees of freedom
 - (B) Probability
 - (C) Predictor size
 - (D) All of the Above

- (E) None of the Above
19. Which model is hierarchical in nature?
- (A) Model 10
 - (B) Model 2
 - (C) Model 6
 - (D) All of the Above
 - (E) None of the Above
20. In Model 10, κ controls:
- (A) Concentration
 - (B) Variance only
 - (C) Sample size
 - (D) All of the Above
 - (E) None of the Above
21. Which PyMC function generates posterior samples?
- (A) pm.Normal()
 - (B) pm.Beta()
 - (C) pm.sample()
 - (D) All of the Above
 - (E) None of the Above
22. In PyMC, the keyword **observed=** is used for:
- (A) Known data
 - (B) Prior definition
 - (C) Deterministic variables
 - (D) All of the Above
 - (E) None of the Above
23. Which distribution is commonly used for probabilities between 0 and 1?
- (A) Beta
 - (B) Normal
 - (C) Poisson
 - (D) All of the Above
 - (E) None of the Above
24. Which model uses Bernoulli likelihood?
- (A) Models 1, 4, and 10
 - (B) Models 2 and 3
 - (C) Models 5 and 7
 - (D) All of the Above
 - (E) None of the Above
25. Which model directly models count-valued observations?

- (A) Model 8
 - (B) Model 3
 - (C) Model 7
 - (D) All of the Above
 - (E) None of the Above
26. In Bayesian inference, priors represent:
- (A) Initial beliefs
 - (B) Observed outcomes
 - (C) Sampling algorithms
 - (D) All of the Above
 - (E) None of the Above
27. Which model uses Student-t likelihood?
- (A) Model 9
 - (B) Model 2
 - (C) Model 4
 - (D) All of the Above
 - (E) None of the Above
28. The function `pm.math.exp()` in Model 8 ensures:
- (A) Positive rate values
 - (B) Binary outcomes
 - (C) Smaller variance
 - (D) All of the Above
 - (E) None of the Above
29. Which model computes a decision boundary?
- (A) Model 4
 - (B) Model 2
 - (C) Model 7
 - (D) All of the Above
 - (E) None of the Above
30. Deterministic variables in PyMC are created using:
- (A) `pm.Deterministic()`
 - (B) `pm.Normal()`
 - (C) `pm.Bernoulli()`
 - (D) All of the Above
 - (E) None of the Above
31. A dataset contains only values 0 and 1 representing success or failure. Which model is most appropriate?
- (A) Model 1

- (B) Model 2
 - (C) Model 8
 - (D) All of the Above
 - (E) None of the Above
32. A researcher wants to model the relationship between house price and house size using a straight-line relationship. Which model is most suitable?
- (A) Model 3
 - (B) Model 8
 - (C) Model 10
 - (D) All of the Above
 - (E) None of the Above
33. A classification problem contains two predictor variables and binary output labels. Which model should be selected?
- (A) Model 6
 - (B) Model 2
 - (C) Model 9
 - (D) All of the Above
 - (E) None of the Above
34. A dataset contains count observations with many zeros. Which model would be most appropriate?
- (A) Model 8
 - (B) Model 5
 - (C) Model 7
 - (D) All of the Above
 - (E) None of the Above
35. A dataset contains severe outliers and heavy-tailed observations. Which model is best suited?
- (A) Model 9
 - (B) Model 2
 - (C) Model 1
 - (D) All of the Above
 - (E) None of the Above

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Instruction for filling the sheet

1. This sheet should not be folded or crushed
2. Use only blue/black ball pen or 2HB pencil
3. Circle should darkened completely and properly
4. Erase marked circle completely for deselect

WRONG METHOD



CORRECT METHOD



NAME :

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Q#	Ans	Q#	Ans	Q#	Ans
1	B	13	A	25	A
2	A	14	A	26	A
3	B	15	A	27	A
4	C	16	A	28	A
5	A	17	A	29	A
6	B	18	A	30	A
7	A	19	A	31	A
8	B	20	A	32	A
9	A	21	C	33	A
10	B	22	A	34	A
11	A	23	A	35	A
12	A	24	A		